

Name: AYUB KIPROTICH MAINA

Msc. Molecular Medicine

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Date of birth: May 3rd, 1987.

Nationality: Kenyan.

Languages: English, Kiswahili.

Career Goal: To improve the quality of life through biomedical research, teaching, sales and marketing or otherwise, by use of the knowledge and skills I possess and those that I am yet to acquire.

ACADEMIC QUALIFICATIONS AND AWARDS

May 2011-November 2017: Jomo Kenyatta University of Agriculture and technology: College of Health Sciences-School of Biomedical Sciences, Kenya Medical Research Institute, MSc. Molecular Medicine.

Thesis title: 'Potential of CD 26/DPPIV usage as a surrogate marker for antiretroviral therapy efficacy in HIV Infected populations'.

2006-2010: Egerton University, Njoro, Kenya BSc. Second class honours (Upper division) in Biomedical Science and technology

2001-2004: Kapsabet Boys High School, Kapsabet. Kenya Certificate of Secondary Education (KCSE)-mean grade A- of 78 points.

2000: Baraka school, Kapsabet. Kenya Certificate of Primary Education (KCPE)-517 Marks out of 700.

PROFESSIONAL EXPERIENCE

March 2025-September 2025: Sales assistant, Inqaba Biotec East Africa Limited. Sales and marketing of products such as Zymo Research, New England Biolabs, Eppendorf, Seegene, Glentham Life Sciences, Scharlab, Genaxy amongst others.

September 2024-February 2025: Part time lecturer, Kaimosi Friends University, School of Health Sciences, teaching Cell Biology to Nursing and Clinical medicine students.

April 2022-December 2023: Teacher, Stanley Samoei Secondary school Kiptenden, Kapsabet.

September 2017-March 2022: Teacher-Baraka School, Kapsabet

Specific duties:

- Giving knowledge and advice to primary school children

January 2016-July 2016: Intern at the Kenya Medical Research Institute

Specific duties:

- Cell culture of peripheral blood mononuclear cells
- Performing p-24 assays to the cells mixed with drugs
- Viral load quantification
- CD 4+ cell counts

May 2015-November 2015: Graduate trainee, McGill University AIDS Centre, Montreal, QC, Canada, working on HIV research in the late Dr. Mark Wainberg's laboratory.

Specific Duties:

- Cell culture of peripheral and cord blood mononuclear cells
- Mixing of drugs in different concentrations
- Exposure of the grown cells to different concentrations of the drugs
- Performing of reverse transcriptase assays

March 2014-April 2015: Intern at the Kenya Medical Research Institute Production Department working on hepatitis research.

Specific duties:

- Arranging Hepatitis B and C samples from different sides of the country
- Performing rapid diagnosis assays to the samples
- Performing ELISAs and PCRs to the samples

June 2013-February 2014: Product specialist for Abbott Diagnostics Division under Philips Healthcare Technologies assisting laboratories using hematology, clinical chemistry and immunoassay medical equipment in calibration, quality control and general use of the equipments.

Specific duties:

- Meeting with prospective clients
- Selling of Abbott diagnostics' medical equipment
- Product specialist for Abbott customers

April-May 2013: Department of Medical Microbiology, University of Manitoba. 2 months training and working in the laboratory on various techniques

Specific Duties:

- Extensive training in regular PCR with gel electrophoresis in the typing of the Interleukin 6 gene polymorphisms in HIV exposed but uninfected (HESN) cohort, between rapid and slow sero-converters.
- Extensive training in Real-Time PCR
- Extensive training of working in Level 2 laboratory with Level 3 specifications
- Online Training on the Pathogen Risk Assessment by Public Health Agency of Canada
- Training in cell culture techniques, like thawing of already frozen PBMCs and re-suspending them in RPMI media with the various PBS-FCS washing steps.
- Extensive training in cell differentiation and cell counts using haemocytometer

- Additional Training in basic flow cytometry techniques
- Attended the weekly departmental presentations on various issues and techniques in medical microbiology.
- Attended the video conferencing session hosted by the University of Manitoba PHD students and including participants from India, Kenya and Colombia on HIV/AIDS situation in the four regions of the world.

May-August 2009: Kenya Medical Research Institute (KEMRI), Nairobi, Kenya (<http://www.kemri.org>) attachment

Specific Duties:

- Induction training in Molecular Biology techniques
- Techniques in the maintenance of *Schistosoma* lifecycle
- Maintenance of mosquitoes and sand fly lifecycles
- Basic training in the use of a FacsCalliber flow cytometry machine in CD4+ cell counts

January-August 2006: Teacher Baraka School

Specific duties

- Giving knowledge to primary school students and advice to improve their outlook in life

PUBLICATION

“The Potential for DPPIV/CD26 usage as a surrogate marker for Antiretroviral Therapy Efficacy in HIV Infected populations.”: Ayub K. Maina, Joseph K. Mbugua, Adriana Trajtmán, Yoav Keynan, Robert W. Omange, Annie Panikulam, Rachel N. Musoke, James Kimotho, Missiani Ochwoto, Daniel Kariuki and Elijah M. Songok. *African Journal of Pharmacology and Therapeutics*. Vol 5 No. 4 pages 263-271. 2016.

“Risk factors of virologic failure and slow response to ART among HIV-infected children and adolescents in Nairobi.”: J. M. Kabogo, S. Gupta, A. K. Maina, M. Ochwoto, R. W. Omange, R. N. Musoke, R. W. Lihana, E. Muniu, F. W. Wamunyokoli, B. Liang and E. M. Songok. *East African Medical Journal*. Vol. 91 No. 4 pages 82-89. May 2017.

DISSERTATION AND THESIS TITLES

Post Graduate: Msc: The potential of CD 26/DPPIV usage as a surrogate marker for antiretroviral therapy efficacy in HIV Infected populations.

Supervisor: Prof. Elijah M Songok

Undergraduate: BSc: Antibiotic resistance in commensal *E.coli* and food borne *Salmonella* in Commercial broiler chicken farms in the Nakuru and Njoro regions of Kenya.

Supervisor: Prof. Anakalo Shitandi

CONFERENCE ATTENDANCE

- Presented my Msc project at the annual HIV/AIDS collaborative group meeting organized by the University of Manitoba and University of Nairobi from 26th to 30th January 2015 in Nairobi.
- Presented my Msc project at the 5th KEMRI Annual Scientific and Health (KASH) Conference, from 4th to 6th February 2015, at the KEMRI Headquarters in Nairobi, Kenya.
- Attended the annual HIV/AIDS collaborative group meeting organized by the University of Manitoba and University of Nairobi from 27th to 31st January 2014 in Nairobi.
- Attended the Egerton University Biomedical Society (EUBIS) conference on various aspects in biomedical research, 2007.

SHORT COURSES

1. Online Training on the Pathogen Risk Assessment by the Public Health Agency of Canada and the containment levels for each risk group, in April 2013 at the University of Manitoba.
2. Professional training by the Rotaract club of Egerton University, 2007 and 2008.

OTHER SKILLS

Packages proficiency:

Ms Word, Ms Excel, Ms PowerPoint, Ms Access.

Operating systems: Linux SUSE 10.1, Windows XP, Millennium, 2000, 98, 95, 7,8.

Internet tools: Microsoft Internet Explorer, Netscape Navigator, Mozilla FireFox, Google chrome, Safari

Task specific packages: Basic knowledge in BLAST.

Statistical analysis: SPSS

TECHNICAL SKILLS

Molecular Biology Techniques

PCR-based techniques

Non-PCR based Techniques: Southern Blotting.

Immunological- Western blots, Enzyme Linked Immunosorbent Assays (ELISAs)

Biochemical techniques

Analytical Chemistry

Spectrophotometry

Microbiology techniques

Bacterial Culture and drug sensitivity testing

Applied microbiology

Diagnostic microbiology

COMMUNITY SERVICE

- Helping Internally Displaced Persons in 2008 at the Nakuru show grounds
- Various Tree planting visits
- Community empowerment in arid areas of Baringo, Turkana and Coastal region of Kenya
- Cleaning of the community resources
- Various walk to raise funds to assist orphaned children.

REFERENCES

Prof. Elijah M. Songok
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Prof. Daniel Kariuki
Department of Biochemistry,
Jomo Kenyatta University of Agriculture and Technology (JKUAT),
Nairobi, Kenya.
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